2167 Local Control, Patterns of Failure, and Visual Acuity following Treatment of Ocular Melanoma Utilizing a ^{125}I Episcleral Plaque: A Single Surgeon Experience

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Purpose/Objective(s): To evaluate clinical outcomes of Choroidal Melanoma (CM) patients treated with ^{125}I episcleral plaque brachytherapy and compare our single surgeon results to the multi-institutional Collaborative Ocular Melanoma Study (COMS).

Materials/Methods: Retrospective review was performed of all CM patients treated with ^{125}I episcleral plaque brachytherapy at Allegheny General Hospital from 6/1996 to 10/2008. All plaques were applied by a single ophthalmologist in accordance with established COMS guidelines.

Results: The records of 35 patients (17 male, 18 female; median age 68 years) were available. The median lesion diameter and elevation was 13mm (range 7 - 16 mm) and 7.5 mm (range 3 - 13.5 mm) respectively, while median plaque size was 17 mm (range 14 - 20 mm). Median dose to the apex was 8609 cGy (range 7500-10500 cGy) at a median dose rate of 92 cGy/hr (range 77 to 140 cGy/hr). The median implant duration was 96 hours (range 71.5-105 hrs.). At a median follow-up of 5 years, 33 patients achieved local control and successful organ preservation, while twenty-three retained useful vision in the treated eye. One patient developed a second ipsilateral CM, which was effectively controlled by repeat ^{125}I episcleral plaque application. Two patients required enucleation at 24 and 30 months post-implant for pain and neovascular glaucoma, respectively. Two patients developed hepatic metastasis at 19 and 21 months post-implant, and are the only mortalities to date.

Conclusions: Our 5-year local control rate (94.3%), distant metastatic rate (5.7%) and mortality with metastatic disease (5.7%) compare favorably to COMS respective 5-year values of 89.7 %, 25%, and 10%. Application of the ^{125}I episcleral plaque by an experienced ophthalmologist cannot be undervalued and may optimize dose delivery.

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2168 Stereotactic Radiosurgery Alone for Patients with 1-4 Radioresistant Brain Metastases

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Purpose/Objective(s): Long-term survivors of brain metastases are at risk for developing neurocognitive deficits after whole-brain radiotherapy (WBRT). Brain metastases from radioresistant histologies are perceived to be less responsive to WBRT compared to other histologies and stereotactic radiosurgery (SRS) may provide better local control. The aim of this study was to examine the outcomes of patients with 1-4 brain metastasis from radioresistant histologies (renal cell carcinoma and melanoma) treated with SRS alone.

Materials/Methods: Exempt review was granted by our cancer hospital institutional review board for the collection of data of patients with radioresistant brain metastases treated with Gamma Knife (GK)-based SRS in our department. In the period of 2000 to 2007, a total of 38 patients (20 male and 18 female) with 1-4 radioresistant brain metastases (66 lesions) were treated with SRS alone. The median age was 55 years (range, 27-81 years). Fourteen and 24 patients had renal cell carcinoma (RCC) and melanoma brain metastases, respectively. Distribution of number of lesions was as follows: one lesion, 22 patients; 2 lesions: 8 patients; 3 lesions, 5 patients; and 4 lesions: 3 patients. Distribution of RTOG recursive partitioning analysis (RPA) classes was as follows: II, 37 patients; and III, 1 patient. Out of 38 patients, 4 had stable extracranial disease.

Results: The median marginal dose was 20 Gy (range, 15-22 Gy). The combined target volume ranged from 0.054 to 15.9 cm^3. The follow-up intervals ranged from 1.3 to 30.9 months (median, 6.1 months). Twenty patients (58%) developed distant brain failure. Nine of the 38 patients were alive at last follow-up. The 3-, 6-, 9-, 12- and, 18-month local control (LC) rates were 87.9%, 81.4%, 67.9%, 67.9%, and 60.3%, respectively. The corresponding free-from-distant-brain failure (FFDBF) rates were 71.3%, 58.1%, 49.8%, 40.2%, and 27.6%. The corresponding progression free survival (PFS) rates were 55.3%, 41.9%, 33%, 23.3%, and 13.3%. The corresponding overall survival (OS) rates were 78.9%, 54.5%, 45.5%, 36.2%, and 24.1% (median OS, 6.5 months). RCC histology was associated with better LC ($p = 0.0055$), although histology did not impact on OS, PFS, or FFDBF. The number of lesions (1 vs. 2-4) did not impact on OS, PFS, or FFDBF.

Conclusions: Although SRS alone could yield reasonable LC in patients with 1-4 radioresistant brain metastases, the risk of distant brain failure was substantial ($\sim$40% and $\sim$60% at 6 and 12 months). RCC histology was associated with better LC although it did not impact OS, PFS, or FFDBF. The number of lesions (1 vs. 2-4) did not impact on OS, PFS, or FFDBF.

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2169 Fractionated Stereotactic Radiotherapy for Intracranial Meningioma: The Long-term Outcomes in Single Institution

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Purpose/Objective(s): For intracranial meningioma, fractionated stereotactic radiotherapy (FSRT) has been expected to increase therapeutic ratio comparing to conventional radiotherapy and to single high dose radiosurgery. We have investigated single institutional long-term outcome of linac-based FSRT in last 10 years.

Materials/Methods: Thirty-four patients with skull base meningiomas who were treated with FSRT were retrospectively reviewed. The age ranged from 14 to 79 years old (median, 58.0). There was 9 male and 25 female. Thirty cases were diagnosed